UNIVERSIDAD DE LAS FUERZAS ARMADAS

INNOVACIÓN PARA LA EXCELENCIA

ESPE

GITHUB:

https://github.com/Yeikolon/Practica01.git

SUBJECT:

Web Fundamentals Systems

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Responsive Adaptation Analysis

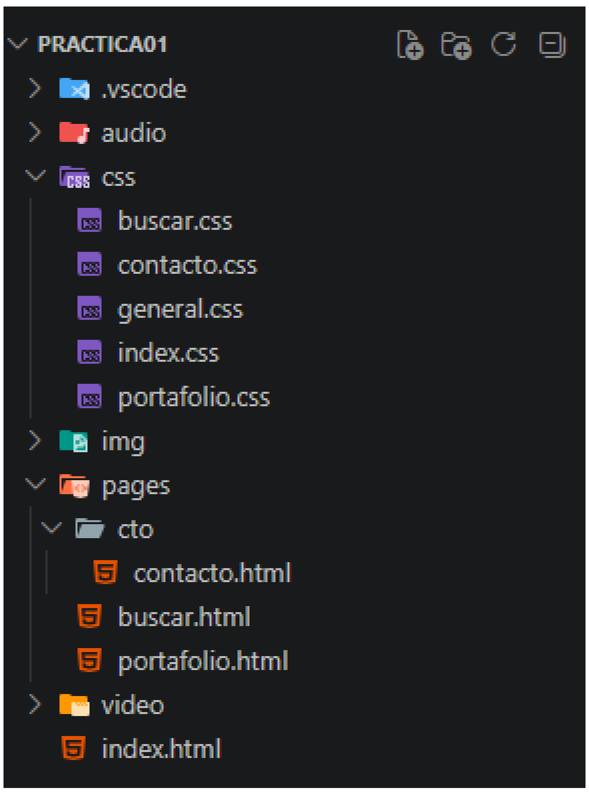
1. Project Overview

Purpose of the website

The purpose of the website is to learn HTML, with each expansion providing new knowledge of tags or in this case CSS.

Pages included in the project

Figure 1: Estructure of the project



Purpose of the new portafolio.html page

The purpose of the portfolio page is to indicate all the most relevant information about the different subjects I am currently taking in my fourth semester.

2. CSS Fundamentals Applied

CSS syntax

```
img{
    width: 100%;
    height: 200px;
    object-fit: cover;
}
```

The syntax is simple: first, you put the element to be modified, then open curly braces, and inside those braces, you put the attribute you want to modify. Finally, you use a colon to indicate the value of that attribute. Remember to put a semicolon at the end of each line.

Inline CSS

```
<h1 style="text-decoration: underline;">Academic Portfolio</h1>
```

This project didn't originally include any inline CSS, but I've temporarily added some for the sake of explanation. Essentially, you need to place it within the element you want to modify. Using the "style" attribute in quotes, you specify all the styles you want to apply. The syntax is like the previous example, but it's usually done on a single line, which makes the code look messy. That's why it's not recommended to use it inline, but the style applied inline will have the highest priority.

Internal CSS

```
<style>
    h1{
        text-decoration: underline;
    }
</style>
```

It has lower priority than the previous example; this one is placed within the same HTML file and only affects that file. It's placed inside the <head> tag, and to use it we use the <style>, which uses the same CSS syntax.

External CSS

```
<link rel="stylesheet" href="../../css/general.css">
```

This has even less priority but is the most organized way; it is done in a separate CSS file, and the way to indicate that we want to use it is with <link> within which we will indicate that the relationship is of type "stylesheet" and the href which is the path where the CSS file is located.

Selectors

```
img{
    width: 100%;
    height: 200px;
    object-fit: cover;
}
```

A selector is simply the element to be modified; in this case, it's img.

Classes

```
.card p{
    text-align: center;
}
```

To make a change to a class, place a period followed by the class name, and the modifications will be made to all elements of that class.

IDs

```
#objetivo {
    background-color: #f7fff4;
}
```

To make a change to an element that has a unique ID, place the number sign (#) followed by the ID name; the change will only be made to the element with that ID.

CSS properties

```
img{
    width: 100%;
    height: 200px;
    object-fit: cover;
}
```

The properties are the attributes to be modified in this case: width, height and object-fit.

3. Visual Design Elements

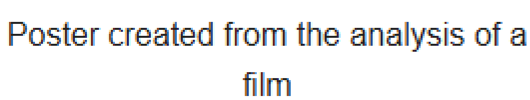
Colors

Figure 2: Color of h2

The color option changes the font color; it's used to prevent everything from being the same color and visually improves the overall look because the colors coordinate.

Fonts

Figure 3: Fonts



I changed the font of the letters; I used it to give the page a more professional design, and that's precisely why it improves the design.

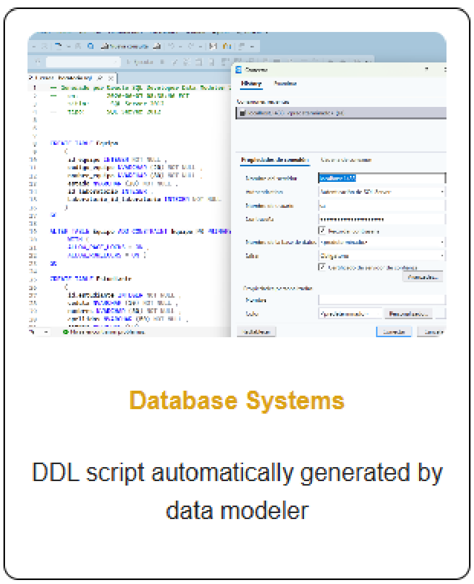
Backgrounds

Figure 4: Background

Background changes the background color of any element; use it to contrast colors and make it look nicer.

Borders

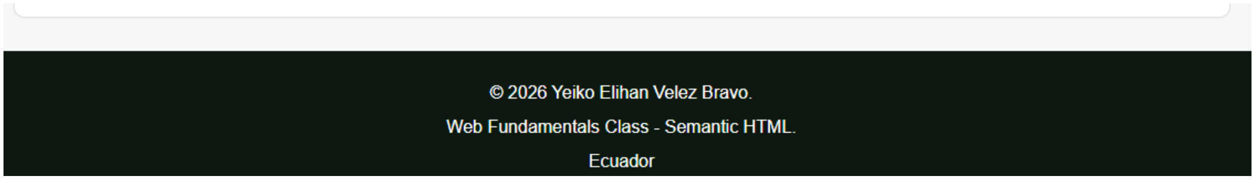
Figure 5: Border of cards



Border allows you to change the border thickness, type, and color. I use it to highlight where each card ends, which improves the design by making the card boundaries clear and organized.

Margins

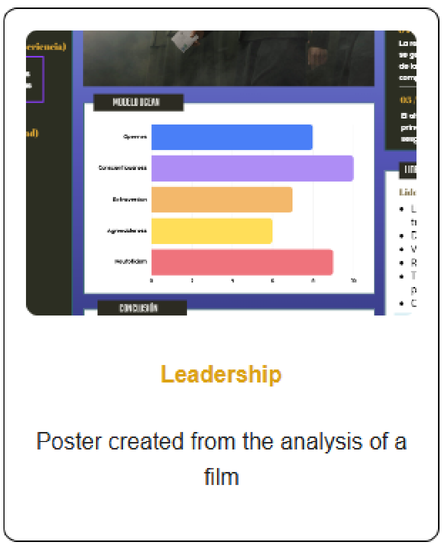
Figure 6: Footer



The margin adds spacing outside the element; I used it to create more space between the footer and the previous section. It helps to make everything look more segmented.

Padding

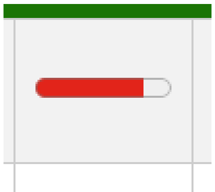
Figure 7: Padding of cards



Padding also creates space, but not outside, rather inside the element itself. I use it so that images don't take up the entire card, and this improves the design because it formats the image.

Width and height

Figure 8: Progress

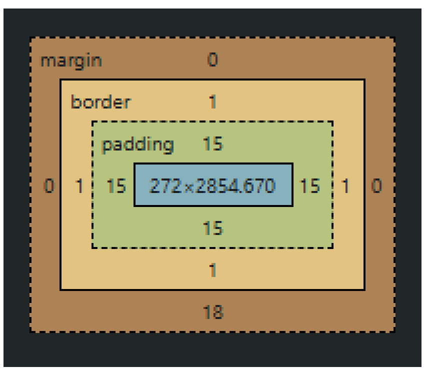


Width and height are used to modify both the height and width of the elements; in this specific case, I used them to give the progress an appropriate size.

4. Layout and Structure

Box Model

Figure 9: Box Model of section



As you can see in the example, we can observe the layers that all the elements have: the margin (the outer layer), the border (the layer surrounding the element), the padding (the layer between the border and the element), and finally, the element itself. Each of these layers can have varying dimensions.

```
* {
    box-sizing: border-box;
}
```

This global configuration ensures that the padding and border do not increase the total width.

Display

```
.container{
    display: flex;
    gap: 20px;
    flex-wrap: wrap;
    justify-content: center;
    height: auto;
    border: 0;
}
```

The display indicates how an element (preferably one to be used as a container) will organize or place the items that are inside it.

Flexbox

```
.container{
    display: flex;
    gap: 20px;
    flex-wrap: wrap;
    justify-content: center;
    height: auto;
    border: 0;
}
```

Continuing the previous example, to create a flexbox display, flex was used. This helps to position the items either horizontally or vertically, which was useful for organizing the cards.

5. Responsive Design Analysis

- **Why are media queries necessary?**

  Media queries are necessary because they allow a website to adapt to different devices without creating separate websites for each device.

- **What changes occur when the screen width is 700px or less?**

  When the screen is reduced to 700 pixels wide or less, the website transforms from a desktop design to one adapted for mobile devices.

- **Why do navigation elements change layout?**

  Navigation elements change layout on smaller screens for usability, accessibility, and user experience reasons.

- **Why do forms use width: 100%?**

  To ensure input fields, selects, textareas, and buttons fill the entire available width of the screen, making them easier to tap, read, and use on small devices.

- **Why does max-width: 100% prevent overflow?**

  Because by setting a width limit, it prevents it from going outside the container it's in.

- **Why box-sizing: border-box is important**?

  Ensures that padding and borders do not increase the overall width or height of an element.

- **How does responsive design improve usability on mobile devices?**

  Responsive design makes websites automatically adapt to the screen size of the device, so mobile users can read, tap, and navigate without zooming or horizontal scrolling.

6. Mobile-First Reflection

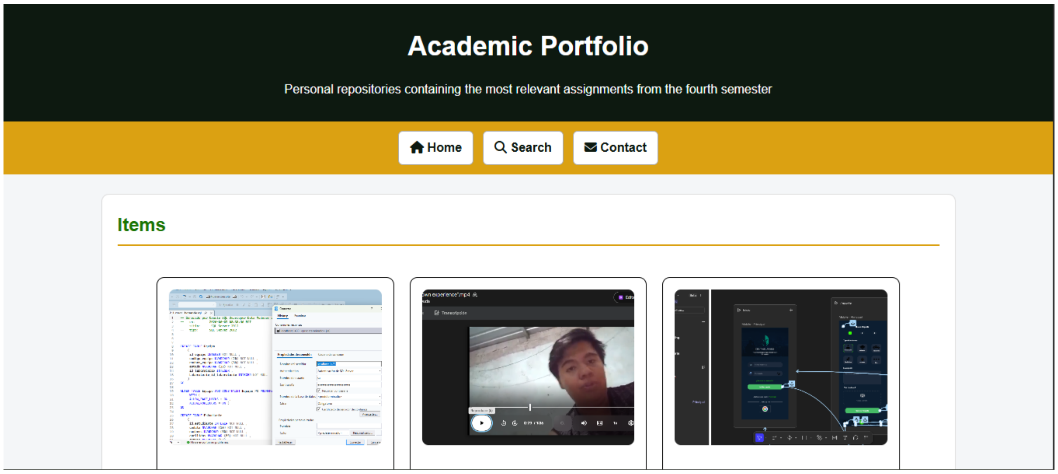
Mobile-first design is the strategy, responsive design is the technique, and a good user experience is the result.

The project uses a desktop-first responsive design, which offers good user experience. However, a mobile-first approach would improve it even further, as it would be optimized first for the most demanding device.

7. Screenshots and Evidence

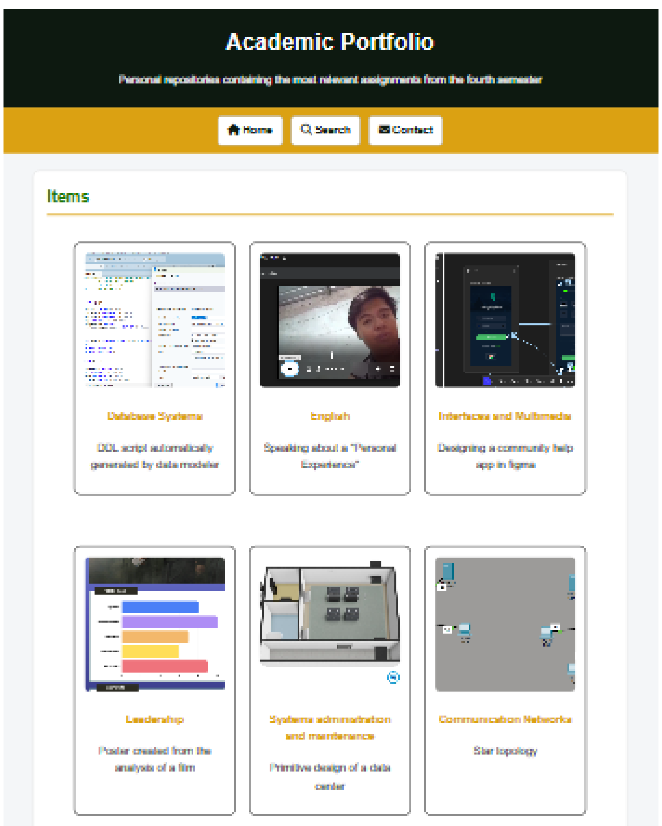
Desktop view

Figure 10: Desktop View



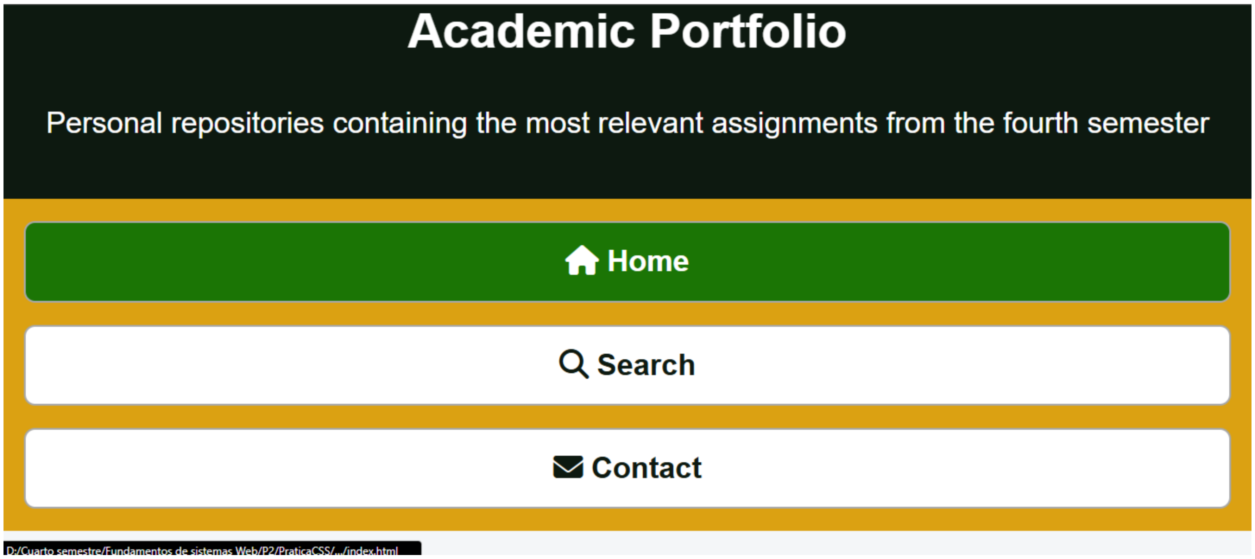
Mobile view

Figure 11: Mobile view



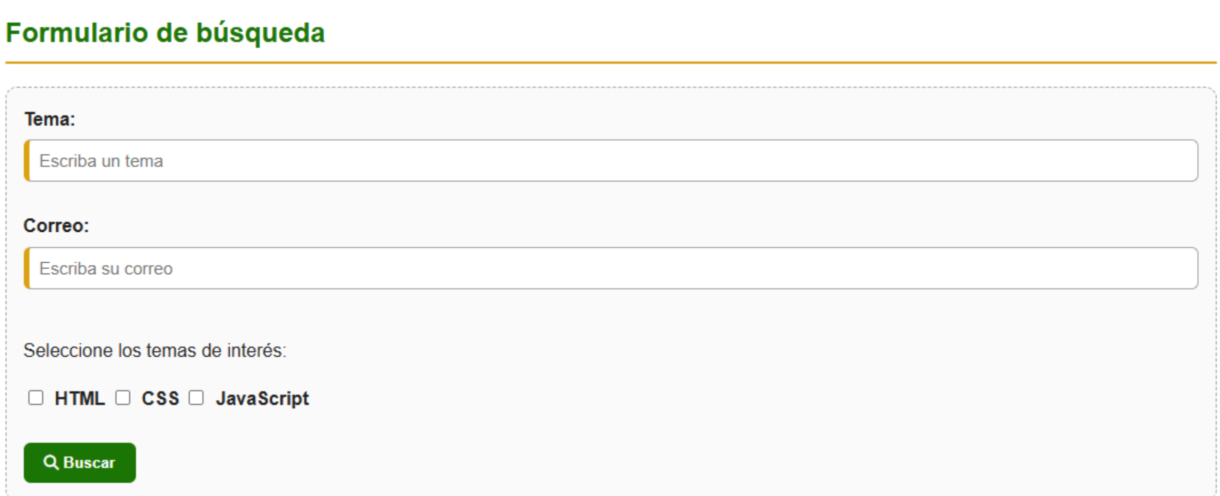
Navigation menu adaptation

Figure 12: Adaption of 200%



Forms

Figure 13: Form



Tables

Figure 14: Table of information

Subject	Information	Progress	Skills	Learning outcomes
Database Systems	Relational design, ER modeling, and SQL		Query optimization and schema design	Design, implement, and manage efficient business databases
English	Technical and professional communication skills		Technical reading, professional writing, and conversation	Collaborate in multicultural teams and comprehend technical documentation
Interfaces and Multimedia	UI/UX design principles for web and mobile		Figma, wireframing, and responsive design	Create intuitive, accessible, and engaging user interfaces
Leadership	Team management, conflict resolution, and ethics		Assertive communication, delegation, and negotiation	Lead high-performance teams to achieve strategic objectives
Systems administration and maintenance	Management of server operating systems		User and permissions management	Ensure infrastructure availability, security, and business continuity
Communication Networks	OSI/TCP-IP models, routing, and addressing		Router and switch configuration, subnet design	Design and implement secure, scalable network architectures
Web Systems Fundamentals	Client-server architecture, HTML, CSS, JavaScript and HTTP concepts		Modern web layout	Develop functional, dynamic, and responsive modern websites

Images

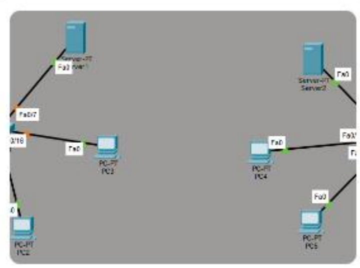
Figure 15: Images of cards

Leadership

Poster created from the analysis of a film

Systems administration and maintenance

Primitive design of a data center



Communication Networks

Star topology

New portafolio.html page

Figure 16: Portafolio Part 1

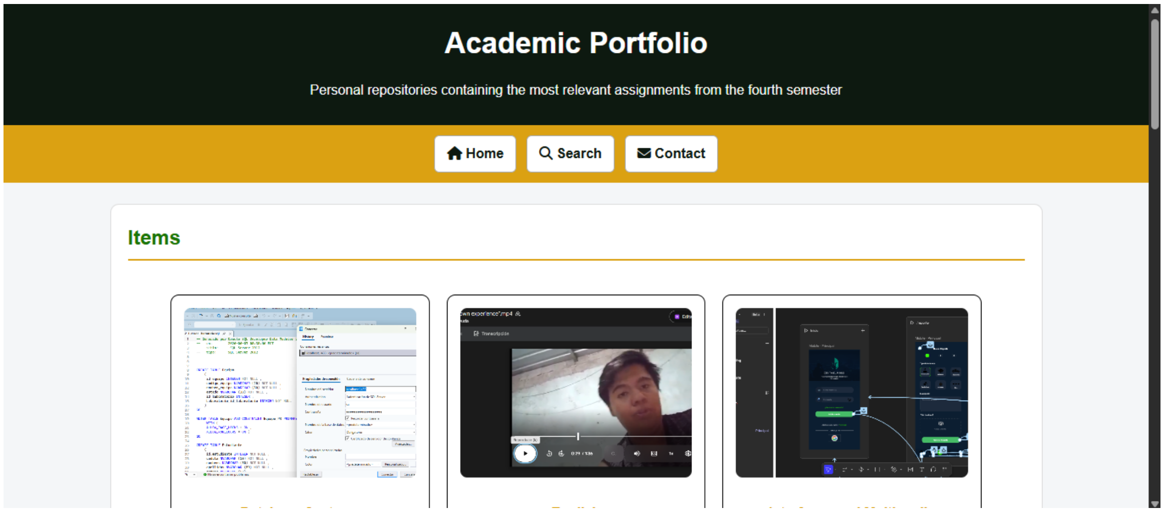


Figure 17: Portafolio Part 2

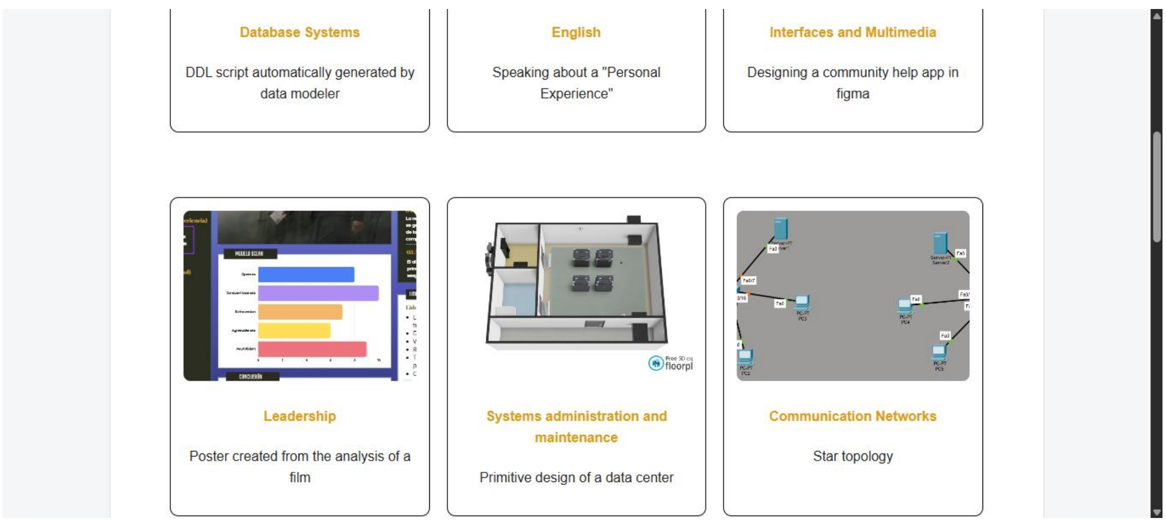


Figure 18: Portafolio Part 3

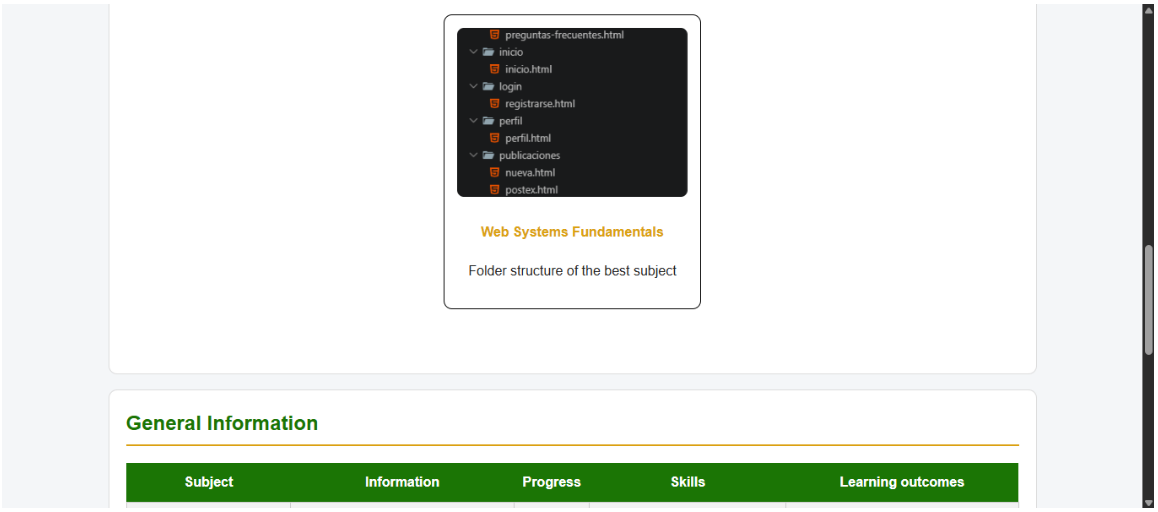
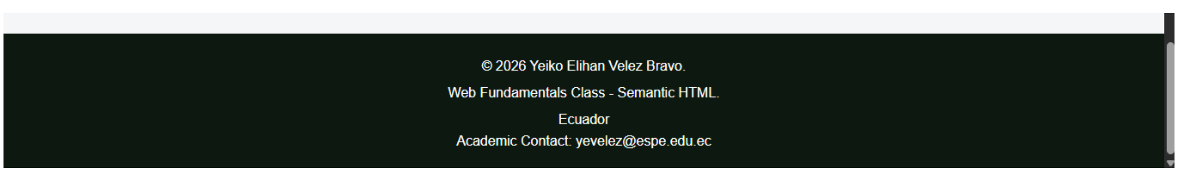


Figure 19: Portafolio Part 4

Database Systems	Relational design, ER modeling, and SQL		Query optimization and schema design	Design, implement, and manage efficient business databases
English	Technical and professional communication skills		Technical reading, professional writing, and conversation	Collaborate in multicultural teams and comprehend technical documentation
Interfaces and Multimedia	UI/UX design principles for web and mobile		Figma, wireframing, and responsive design	Create intuitive, accessible, and engaging user interfaces
Leadership	Team management, conflict resolution, and ethics		Assertive communication, delegation, and negotiation	Lead high-performance teams to achieve strategic objectives
Systems administration and maintenance	Management of server operating systems		User and permissions management	Ensure infrastructure availability, security, and business continuity
Communication Networks	OSI/TCP-IP models, routing, and addressing		Router and switch configuration, subnet design	Design and implement secure, scalable network architectures
Web Systems Fundamentals	Client-server architecture, HTML, CSS, JavaScript and HTTP concepts		Modern web layout	Develop functional, dynamic, and responsive modern websites

Figure 20: Portafolio Part 5



8. Personal Reflection

What was learned about responsive web design

Through this project, I learned that responsive web design is not just about making a website "look good" on phones it's about fundamentally changing how users interact with the content based on their device.

Challenges encountered

- The 6 cards became jumbled when the screen was reduced.
- Some images appeared distorted or cropped.

Improvements that could be made in future versions

Create responsive cards that change from 3 columns to 2 columns to 1 column depending on the screen